Identifiability Limits of Cusp Models for Overload in Wearable Physiology, and a Ground-Truth Failure of Rolling-Window Early-Warning Estimators

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Abstract—Catastrophe models promise a generative account of overload: a latent load coordinate relaxing in a quartic potential, functional states as metastable basins, sudden onset and slow recovery as consequences rather than observations. We ask what such a model can actually recover from recordings of the length wearable corpora supply, and answer with a Monte Carlo study over 1.2×10^4 simulated series. Only the relaxation rate and the diffusion coefficient are identified. The parameters carrying the geometry are ratios whose denominator is not bounded away from zero, so length buys their ordering and never their value: rank correlation for the splitting factor reaches 0.88 at $n = 1000$ while its linear correlation stays at 0.10. That bound holds for any cusp analysis of data of this kind, ours included, and it is the paper's primary result. We then show that the rolling-window variance and autocorrelation indicators the early-warning literature relies on return exponents of the wrong sign on data simulated from a model that does contain a fold, wrong in 99% of replicates at a 30-sample window, while the exponent computed from the true curvature recovers correctly. The failure is structural: the correlation time diverges at the fold while the window stays fixed. We withdraw our own early-warning test rather than report it. Applying the one prediction that remains identifiable to 147 recordings from 40 people in three open corpora returns a surrogate-calibrated null, with power characterised so that this excludes strong and moderate dynamics rather than reporting an absence. Along the way we measure an eightfold size inflation of the nominal test on near-unit-root electrodermal data. The work is motivated by sensory and executive overload in autistic adults; none are analysed here, and a pre-registered validation is specified.

Index Terms—Critical transitions, cusp catastrophe, hysteresis, early-warning signals, identifiability, electrodermal activity, wearable sensing, sensory overload.

I. Introduction

Autistic adults describe functional breakdown under combined sensory and cognitive demand in consistent terms: it arrives suddenly, and it clears slowly [1]. Adult cohort studies link sensory atypicality to executive difficulty [2], and meta-analysis finds executive dysfunction across the spectrum, though with smaller effect sizes in adult samples than in child ones [3]. The phenomenon is not in doubt. What is missing is a formal account of the dynamics.

Existing models do not supply one. Computational work on autism is dominated by Bayesian and predictive-coding framings whose empirical support a ten-year review found mixed and methodologically inconsistent [4], and the most recent dynamical treatment of symptom trajectories is avowedly theoretical, collecting no data [5]. Applied state-transition work assigns states from experimental condition labels and counts a transition matrix. That is close to circular: the recovered matrix re-describes the block structure of the protocol, and yields nothing observation could contradict.

We take a different route. Instead of positing states and fitting transitions between them, we posit a one-dimensional stochastic relaxation in a cusp potential and let the phenomenology follow. The load coordinate x_t sits in $V(x; a, b) = \frac{1}{4}x^4 - \frac{1}{2}ax^2 - bx$, where the normal factor b_t carries momentary demand and the splitting factor a_t carries depleted regulatory capacity. The substantive claim is that capacity does not add to load. It changes the geometry. Below a critical a the load tracks demand and returns; above it a second attractor appears, bringing a fold, a hysteresis loop and critical slowing down. Sudden onset and slow recovery stop being observations to report and become consequences to check: numerical exponents that data can reject, since the geometry is closed-form. That is the promise. This paper asks the prior question, and the answer is what it contributes: before asking whether overload is a cusp, one has to ask which parameters of such a model a wearable recording can support. The answer is two, and the two carrying the geometry are not among them. That bound is a property of the estimator, not of our corpora, so it constrains any study of this kind. We show alongside it that the rolling-window indicators such a study would reach for report the wrong sign on data that provably contains a fold. Both results are portable, and both stand independently of the negative empirical result we report on real recordings.

Contributions.

- An identifiability bound on cusp models of wearable physiology (Sec. IV): a 1.2×10^4-replicate study showing the geometric parameters are ratios with an unbounded denominator, so length recovers their ordering and never their value. It binds any study of this kind, ours included.

- A ground-truth failure of the standard early-warning estimators (Sec. V): rolling variance and lag-1 autocorrelation report the wrong sign on data containing a genuine fold, with the mechanism identified, plus the eightfold size inflation of the nominal test on near-unit-root series.
- The formulation the bound is stated for (Sec. II): a cusp model whose five states are geometric regions rather than labels and whose 5×5 kernel follows from nine parameters, fitted by exact profiled least squares.
- A calibrated null on 147 recordings, powered so the result excludes strong and moderate dynamics rather than reporting an absence.

Motivation and scope. The programme this work belongs to targets autistic adults, which is why the phenomenology above is the phenomenology of overload. No public wearable dataset of that group exists at the temporal density the model needs [6], [7], so none are analysed here and the paper is not titled after them. That costs the results nothing, because the results are about estimators. General stress corpora carry the same sensor, sampling rate and recording lengths an autism study would, so a bound measured here transfers to that study directly. No claim about the population transfers, and we make none.

II. The Cusp–Hysteretic Model

A. Fast dynamics and a slow debt variable

The latent load obeys a gradient relaxation with additive noise,

$$\mathrm{d}x_t = \lambda\big[-x_t^3 + a_t x_t + b_t\big]\mathrm{d}t + \sigma\,\mathrm{d}W_t, \tag{1}$$

with control parameters linked to observed drivers and to a slow variable A_t that accumulates recovery debt:

$$b_t = \beta_0 + \beta_S S_t + \beta_T T_t + \beta_U U_t, \quad a_t = \alpha_0 + \alpha_A A_t, \tag{2}$$

$$\mathrm{d}A_t = \varepsilon[\mathrm{sp}(x_t - x_{\mathrm{ref}}) - A_t]\mathrm{d}t, \qquad \varepsilon \ll 1. \tag{3}$$

Here sp is a softplus, S_t, T_t and U_t are sensory, task-switch and uncertainty drivers, and λ sets the timescale without touching the geometry. Load above baseline accrues debt, debt raises a_t, and a higher a_t widens the hysteresis loop, so one episode makes the next easier. That is a mechanistic reading of the refractory period qualitative accounts describe.

B. Folds and two scaling laws

Equilibria solve $x^3 - ax - b = 0$, whose discriminant $4a^3 - 27b^2$ is positive exactly when two stable basins coexist. Folds sit where $V''(x) = 0$, at $x_\pm = \pm\sqrt{a/3}$, giving critical drives $\pm b_c(a)$ with $b_c(a) = \frac{2}{3\sqrt{3}}a^{3/2}$.

Proposition 1 (Hysteresis width, P1). For $a > 0$ the system jumps to the upper branch at $b = +b_c$ and returns at $b = -b_c$, so the loop has width $\Delta b(a) = \frac{4}{3\sqrt{3}}\,a^{3/2}$.

Linearising (1) about a stable equilibrium gives a relaxation rate $\lambda_{\mathrm{rel}} = 3x^{*2} - a$ that vanishes at a fold, so critical slowing down is a theorem here rather than an assumption. With $\mu = b_c - b$ the distance to the fold, the normal form gives $\lambda_{\mathrm{rel}} \propto \mu^{1/2}$, hence

Proposition 2 (Early-warning exponents, P2). $\mathrm{Var}(x) \propto \mu^{-1/2}$ and $-\log \mathrm{AC1} \propto \mu^{+1/2}$.

These are point hypotheses about exponents, differing in kind from the usual directional claim that an indicator rises before a transition, which the sceptical literature shows arises spuriously on almost any autocorrelated series [8], [9].

C. States and transition kernel

Crossing basin membership with the bistability condition partitions the (a, b) plane into five regions (Fig. 1b): calm and focused in the lower basin when monostable, stuck in the lower basin while an overloaded attractor coexists, overloaded in the upper basin, and recovering there once demand has fallen below the level that would have put the system there. The last two matter most: qualitative work uses both words freely, and to our knowledge neither has previously had a precise mathematical meaning. Between-basin moves are noise-induced escapes with Kramers rate $r = \frac{1}{2\pi}\sqrt{|V''(x^*)V''(x_s)|}\,\exp(-2\Delta V/\sigma^2)$ [10], so the kernel is a deterministic function of (a_t, b_t, σ): twenty counted probabilities replaced by nine interpretable parameters that move with demand.

III. Materials and Methods

A. Corpora

Three open corpora, each from its primary repository rather than a mirror, span a gradient of experimental control: WESAD, 15 subjects in a laboratory protocol [11]; Wearable Exam Stress, 10 students across 30 examinations [12]; and Nurse Stress, 15 nurses across roughly 1250 h of hospital shifts [13]. All supply Empatica E4 electrodermal activity, blood volume pulse, skin temperature and accelerometry. Windowing and quality screening leave 147 recordings from 40 people, median 194 windows each.

B. Sensor assignment and a negative control

Read the load coordinate and its drivers off the same sensor and the model stops being falsifiable, since shared measurement noise alone manufactures the coupling; we therefore assign them to disjoint sensors. Which signal carries the load is settled by measurement, not preference: the coordinate of a slow relaxation process must itself be slow. At a 30 s window tonic electrodermal activity has lag-1 autocorrelation 0.98, while a cardiac index from heart rate and RMSSD has 0.19, RMSSD over so few beats being dominated by beat-detection error. Configuration B (tonic EDA) is therefore primary and C (skin temperature) a robustness check. We keep A, the cardiac index, as a negative control: a working pipeline should find nothing

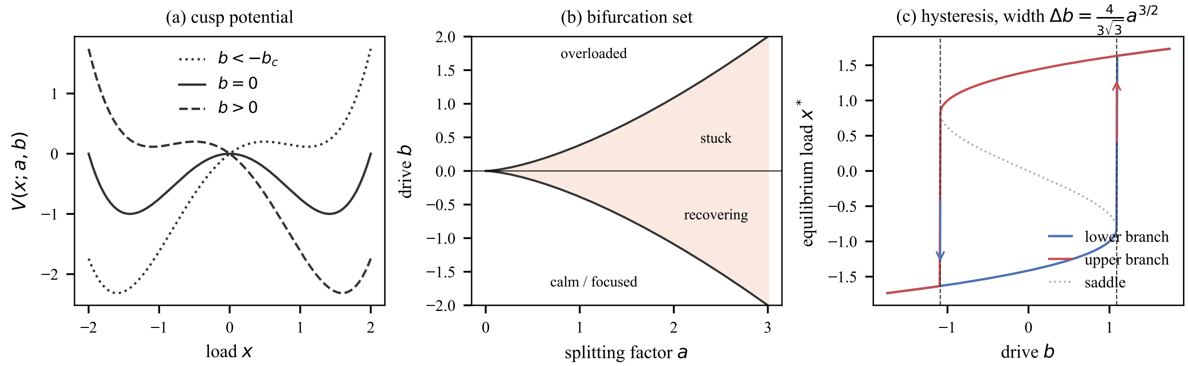

Fig. 1: The geometry from which everything else follows. (a) The cusp potential at three levels of drive b. (b) The bifurcation set in the (a, b) plane, with the five functional states appearing as regions rather than as assigned labels. (c) The hysteresis loop, whose width is fixed by Prop. 1 rather than merely being positive.

in it. Both comparisons are internal to this paper. The cardiac index is a reference channel used to justify a sensor assignment, not a competing method. No published system is claimed to be outperformed in Sec. V, where every baseline is one of our own nested or competing models.

C. Estimation

Taking the Euler map of (1) as the model, rather than as an approximation to it, makes the transition density exactly Gaussian and removes discretisation bias. Expanding the drift, the mean increment is linear in a reparameterised coefficient vector,

$$\Delta x_t = \theta_1(-x_t^3) + \theta_2 x_t + \theta_3 A_t x_t + \theta_4 + \theta_5 S_t + \theta_6 T_t + \theta_7 U_t + \eta_t, \tag{4}$$

with $\theta_1 = \lambda$, $\theta_2 = \lambda\alpha_0$, $\theta_3 = \lambda\alpha_A$ and $\theta_{4:7} = \lambda\beta_{0:U}$. Since A_t depends only on ε, the fit is ordinary least squares profiled over one scalar: exact, no bounds, no starting values, no local optima. We impose $\theta_1 \geq 0$, since a negative value puts the system outside the model rather than merely fitting it badly, and we keep units where that constraint binds so no reported proportion is conditioned on its own outcome.

D. Nulls

The nominal t-test on θ_1 is unusable here. Tonic EDA is close to a unit-root process, and regressing on an integrated series inflates significance in the classical way. We calibrate every test against surrogate ensembles: random-walk surrogates matched on length, increment variance and marginal scale, and amplitude-adjusted Fourier surrogates (IAAFT [14]) that preserve spectrum and marginal, so only nonlinearity can drive a rejection. Five competing models are scored by held-out one-step predictive log-density: homogeneous Markov, covariate logistic, Gaussian HMM, the nested Ornstein–Uhlenbeck case, and gradient boosting.

E. Settings

Windows are 30 s without overlap, swept to 120 s as a robustness check. Calibrated tests draw 200 surrogates, 100 in the power and persistence sweeps; IAAFT runs 200 iterations. The Monte Carlo grid uses $n \in \{100, 150, 200, 300, 500, 1000\}$ at 2000 replicates per cell for recovery (1.2×10^4 series), 5000 for size (30,000 nominal, 7,500 calibrated) and 100 for power; the early-warning validation covers 90 configurations (five window widths, two detrending choices, three lengths to $n = 20{,}000$) at 500 each. Counts were fixed in advance to hold the Monte Carlo standard error on a proportion below 0.01 for size and power. The exception is the AR(1) persistence study at 40 replicates per point, which is why it is quoted as a bound. All of it is seeded from one master seed and runs on one machine, ten workers on a six-core, twelve-thread CPU, about seven hours for a full replication.

IV. What This Data Can Identify

Before testing anything we asked which parameters are recoverable at the series lengths on offer. Not a formality: it turned out to organise the rest of the paper. We simulated with randomised true parameters and refitted, 2000 replicates at each of six lengths, then compared true against estimated values (Table I, Fig. 2a).

Two correlations are reported, because they answer different questions and the gap between them is the finding. Spearman's ρ asks whether the ordering survives; Pearson's r asks whether the number can be read off. For λ and σ they agree, and both are high from $n = 100$ up. Everywhere else they diverge.

The splitting factor carries the geometry, so it is the case that matters. Its rank recovery climbs with length, $\rho = 0.33$ at $n = 100$ to 0.88 at $n = 1000$. Its value recovery does not climb at all: $r = 0.15$ at $n = 200$, $r = 0.10$ at $n = 1000$. There $\hat{\alpha}_0$ spans $[-140, 3.3]$ against a 99th percentile of 2.0, so one blow-up in two thousand replicates destroys the linear correlation. That is the model's own arithmetic, not a quirk of the simulation.

The consequence is sharp, and it binds any study of this kind rather than only ours. Recovered geometry is built from the ratios $a = \theta_2/\theta_1$ and $b = \theta_4/\theta_1$, and nothing bounds θ_1 away from zero. Both therefore inflate without

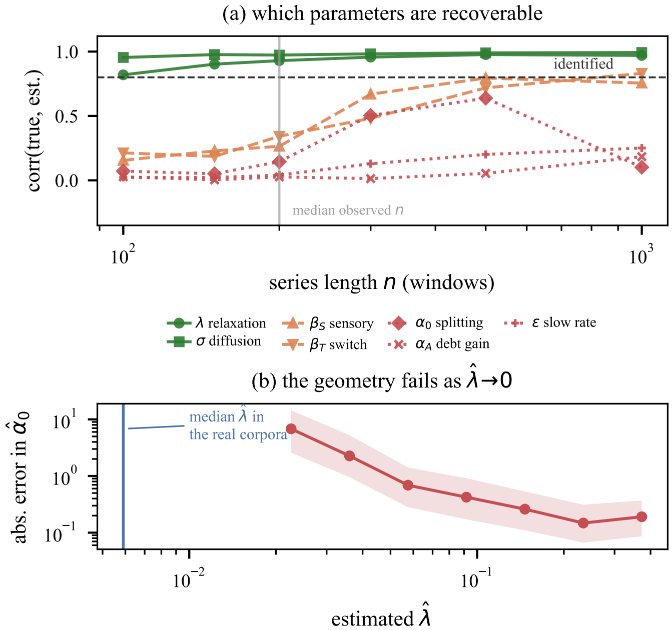

Fig. 2: (a) Rank recovery against series length, 2000 replicates per point. (b) The mechanism: error in the recovered splitting factor rises as $\hat{\lambda}$ falls, because $a = \theta_2/\theta_1$. The median $\hat{\lambda}$ in the real corpora lies below the whole simulated range.

TABLE I: Parameter recovery, 2000 replicates per cell. ρ (Spearman) asks whether the ordering is recovered, r (Pearson) whether the value is. Only λ and σ score well on both. For α_0 they diverge and stay diverged.

	Spearman ρ		Pearson r	
Parameter	n=200	n=1000	n=200	n=1000
λ (relaxation)	**0.94**	**0.98**	**0.93**	**0.97**
σ (diffusion)	**0.99**	**1.00**	**0.97**	**0.99**
β_S (sensory drive)	0.64	0.91	0.26	0.76
β_T (task switch)	0.58	0.88	0.34	0.83
α_0 (splitting)	0.57	0.88	0.15	0.10
α_A (debt gain)	0.32	0.72	0.03	0.18
ε (slow rate)	0.10	0.44	0.04	0.25

bound as $\hat{\lambda}$ falls (Fig. 2b). That explains the pattern in Table I exactly: dividing by a small number preserves order, which is why ρ recovers, and destroys scale, which is why r does not.

The real fits behave the same way. Among units where the constraint does not bind, the median $\hat{\lambda}$ is 0.0059 and the median recovered $\hat{\alpha}_0$ is 12, putting attractors near ± 2 on a coordinate standardised to unit variance. These are not weakly bistable systems. They are systems with no restoring cubic term whatever, whose apparent geometry is an artefact of the division. That median $\hat{\lambda}$ also falls below the entire range our simulation covers, so the recordings sit further into the non-identified regime than any of the 1.2×10^4 synthetic series used to map it.

The practical reading is narrower than "collect more data": length buys a defensible ranking by α_0, never a

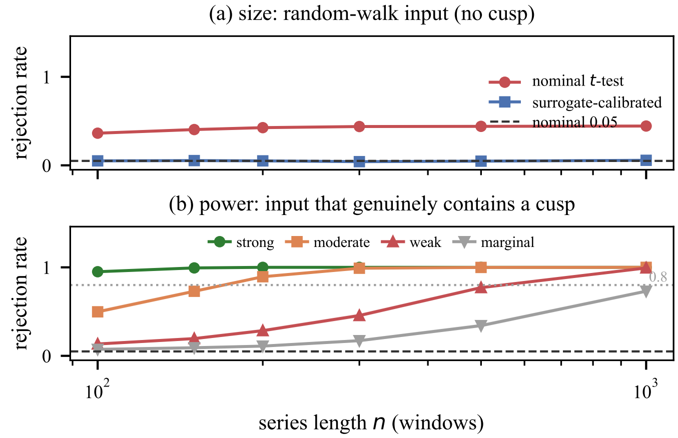

Fig. 3: (a) Size on random-walk input: the nominal test rejects at roughly 0.42 at every length, surrogate calibration restores nominal size. (b) Power on input that genuinely contains a cusp. Bands are Wilson intervals.

reportable value.

One of the five pre-stated predictions survives as evaluable here: the existence of the restoring cubic term, governed by θ_1, the only parameter recovered on both measures. P1 needs the ratio of two badly identified quantities and P2 the distance to the fold, while P3 through P5 rest on ε and α_A. We report that structure rather than four numbers dressed up as tests.

V. Results

A. The nominal test is invalid; the calibrated one is powered

Over 30,000 random-walk replicates spanning six lengths, the nominal one-sided t-test on θ_1 rejects 41.9% of the time rather than 5%. The inflation does not shrink with sample size. It grows, from 36.3% (95% CI $[35.0, 37.7]$) at $n = 100$ to 44.4% ($[43.0, 45.8]$) at $n = 1000$. That is the signature of spurious regression on an integrated series, not a small-sample artefact. Surrogate calibration restores nominal size across the same grid: 5.0% pooled over 7,500 replicates, every per-length interval covering 0.05. Calibration costs power, so we quote power for the test actually used. At $n = 200$, over 2000 replicates per cell, it is 1.00 for strong cusp dynamics, 0.89 for moderate and 0.28 for weak; by $n = 1000$ it is 1.00 for strong and moderate and 0.99 for weak (Fig. 3). The null below therefore excludes strong and moderate dynamics. It does not exclude a faint fold.

B. The early-warning estimator fails on ground truth

Proposition 2 describes the system. Testing it needs an estimator, and we checked ours before trusting it. On data simulated from a model that genuinely contains a fold, the exponent computed from the true equilibrium curvature comes back at +0.565 against a predicted +0.5: theory and implementation agree. The rolling-window indicators the early-warning literature relies on [15], [16] do not. We

TABLE II: Recovering the exponent from ground-truth cusp data, 1500 replicates per row. Theory does; the rolling estimators invert at short windows and collapse to zero at long ones.

Estimator	Pred.	Detr.	Win.	Median	Wrong sign
theory $\lambda_{\mathrm{rel}}(\mu)$	$+0.5$	—	—	$\mathbf{+0.565}$	—
$-\log$AC1	$+0.5$	no	30	-0.216	95%
$-\log$AC1	$+0.5$	no	240	$+0.026$	32%
variance	-0.5	no	30	$+0.307$	99%
variance	-0.5	no	240	-0.014	40%

swept five window lengths, two detrending choices and three series lengths to $n = 20{,}000$: ninety configurations, 500 replicates each. The failure has two regimes (Table II). At windows of 20 to 120 the sign is systematically inverted, and near-totally so: 99% of replicates carry the wrong sign for variance at window 30, and some configurations invert in all 500. At window 240 both indicators instead collapse onto zero, medians between -0.018 and $+0.027$, the sign a coin flip. Nowhere on the grid does an estimate come near the predicted magnitude.

This is structural, and Sec. II says why. The relaxation rate vanishes as $\lambda_{\mathrm{rel}} \propto \mu^{1/2}$, so the correlation time diverges as $\mu^{-1/2}$ while the window stays fixed: near the fold the window is shorter than the process it measures, which is where Table II shows the inversion. Widening it reaches the other end of the same bias, averaging over the approach itself (the window-240 rows), and detrending removes the low-frequency power carrying what survives. We withdraw our own P2 test as uninterpretable.

The claim we draw is narrow, and worth stating at its own size. It covers fixed-width rolling variance and lag-1 autocorrelation with and without Gaussian detrending, which is what the applied literature mostly runs, and says nothing about adaptive-bandwidth or spectral estimators. It indicts those estimators, not early-warning theory: the exponent from the true curvature recovers on the same data. And it sharpens an existing scepticism [8], [9] by supplying a case where ground truth is known, instead of establishing one. Within that scope the caution is firm: a directional early-warning result obtained this way is weak evidence either way.

C. The identifiable prediction is not supported

Under the calibrated test the restoring cubic term is significant in 2.7% of the 147 units against a nominal 5%: 0.0% for WESAD ($n = 15$), 0.0% for the examinations ($n = 30$), 3.9% for the nurse sessions ($n = 102$). Since those 102 units come from 15 nurses we also weight one vote per person, giving 1.5% (Wilson $[0.0, 3.1]$%), with 10% of people showing at least one significant session. That is the most generous reading available, and still what chance produces across several sessions each.

Held-out prediction agrees from another direction. The full model loses to its own nested monostable special case: it wins on 33.8% of units, median difference -0.25 nats per step, Wilcoxon $p = 8 \times 10^{-5}$ for the simpler model. Since the monostable model is this model with the cubic term switched off, that is the cleanest available statement that the term does no work. Prospective onset prediction gives AUC 0.50 to 0.46 at lead times of 3 to 12 windows. Chance.

The model has three separable components: the cubic restoring term, the debt variable and the driver set. That comparison isolates the cubic term, on which the paper's claim rests. We do not ablate the other two in turn, because Sec. IV shows α_A and ε are not recovered at these lengths, and removing a component whose parameters are unidentified attributes a change in fit to something the data cannot resolve.

D. Structure without fold structure

The null is specific, and calling it noise would be wrong. Two of the three calibrated statistics do exceed their surrogate nulls: the likelihood ratio against the monostable restriction at 40.8% of units, marginal bimodality at 49.7%. Only the statistic isolating the cubic term sits at chance. A bimodal marginal with no cubic drift is precisely the configuration in which a study reporting distributional evidence alone concludes the opposite of the truth.

Robustness is mixed, and we report it that way. Configuration B sits at nominal under both the random-walk (0.046) and the harder IAAFT (0.069) ensembles, and holds across 30 s to 120 s windows. A and C are elevated, at 0.11 to 0.16. We tested three explanations for that. All three failed. IAAFT did not bring C down. Clipping rates run opposite to the effect. And an AR(1) size study, ϕ from 0 to 1, found no persistence at which the calibrated test over-rejects, though at 40 replicates per point it would miss an inflation below roughly 0.15. What is left is nonlinear structure the surrogates cannot reproduce. But A is the negative control, built on a coordinate already shown to be noise-dominated, so the parsimonious reading is motion and peak-detection artefact rather than a fold. A test that fires on the negative control tells us significance does not mean cusp there, which is why the conclusion rests on B.

VI. Discussion and Conclusion

Modelling overload as motion in a cusp potential trades a descriptive account for a generative one whose consequences compute in advance. States become basins, the kernel a function of nine parameters, sudden onset and delayed recovery point hypotheses. Precision of that kind costs something: the model can lose. Here it did, twice over and in two different ways. The geometry is not recoverable at these lengths, and the one component that is recoverable, the restoring cubic, is not supported above a calibrated null and is beaten by its own removal on held-out density. Bimodality and a likelihood-ratio excess do survive their surrogates, but a bimodal marginal with no cubic drift is structure of some other kind, and we do not read it as a fold.

The identifiability result is the durable contribution here, and it is a negative methodological one: cusp geometry recovered from short wearable series is largely an artefact of ratios involving a poorly constrained relaxation rate, and rolling early-warning indicators can invert even when a fold is present. Every attempt to fit cusp geometry to wearable physiology inherits the same ratio structure, and the remedy depends on what is wanted. Ranking participants by α_0 needs length, order $n \gtrsim 500$ windows. Reporting a value needs a reparameterisation estimating a and b directly: over the six lengths and 1.2×10^4 series of Sec. IV length alone never fixed it, with r for α_0 at 0.15 for $n = 200$ and 0.10 for $n = 1000$. A splitting factor of 12 from a 200-window recording is not a finding, and the arithmetic producing it is not specific to our implementation.

Limitations. No autistic participants are analysed, so the motivating claim is untested rather than refuted. Tonic EDA is a peripheral, artefact-prone proxy. Weak cusp dynamics stay compatible with everything here. The estimator critique is bounded as stated in Sec. V. Window width we swept; the softplus form of debt accumulation, the driver set (S, T, U) and the decision to treat the Euler map as the model rather than an approximation to it we did not, and each could move the fits. And longer records rescue rankings, not values; not even every parameter in rank, since ε reaches only $\rho = 0.44$ at $n = 1000$.

Toward validation. We pre-specify the study that would test the mechanism in its intended population: 28 days of ecological momentary assessment with continuous wrist physiology in autistic adults and a matched comparison group, using the same estimator, nulls and falsification criteria. The primary hypothesis is a between-group difference in α_0, better powered than the within-group detection attempted here, at $n \gtrsim 500$ windows per participant. The debt mechanism needs more than length: with α_A and ε unrecoverable from passive data at any duration tested, P3 to P5 need either experimental control over recovery intervals or a hierarchical fit pooling them across participants. A second well-powered null under those conditions would be evidence that overload is not a cusp catastrophe at physiological timescales.

Code and Data Availability

The estimator, the analysis pipeline, the Monte Carlo suite, the figure scripts and this manuscript are released under the MIT licence at https://github.com/n-3-0-l-d-3-v/overload-cusp-catastrophe, together with a provenance table recording the replication count behind every number reported. The three corpora are third-party open datasets and are not redistributed: WESAD from the UCI Machine Learning Repository (ID 465), and Wearable Exam Stress and Nurse Stress from PhysioNet. Each was obtained programmatically from its primary repository under that repository's own terms, with the exact URLs, download dates and byte counts recorded in the artefact.

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